

Expt8:

Implement a web application based on the MVC design pattern, to create an Employee Registration module using JSP, Servlet, JDBC, and MySQL database.

AIM:

To implement a web application based on the MVC design pattern with data base.

ALGORITHM

Step 1 : create a database named "employee" in MySQL. Let's create an *employee1* table using below DDL script:

```
CREATETABLEemployee1
(idint,first_namevarchar(20),last_namevarchar(20),usernamevarchar(250),passwordvarchar(20)
,addressvarchar(45),contactvarchar(45));
```

Step 2: Create the *Employee* class which will act as our **Model** class.

Step 3: *EmployeeDao* class that contains JDBC code to connect with the MySQL database.

Step 4: create an *EmployeeServlet* class to process HTTP request parameters and redirect to the appropriate JSP page after request data stored in the database:

Step 5: design an employee registration HTML form *employeeeregister.jsp*

Step 6:After an employee successfully registered then *employeeedetails.jsp* show a successful message on screen

PROGRAM:

Employee.java

package com.pu;

import java.io.*;

```
publicclass Employee {
    privateintid;
    private String firstName;
    private String lastName;
    private String username;
    private String password;
    private String address;
    private String contact;

    public int getId() {
        return id;
    }

    public void setId(intid) {
        this.id = id;
    }

    public String getFirstName() {
        return firstName;
    }

    public void setFirstName(String firstName) {
```

```

        this.firstName = firstName;
    }
    public String getLastName() {
        return lastName;
    }
    public void setLastName(String lastName) {
        this.lastName = lastName;
    }
    public String getUsername() {
        return username;
    }
    public void setUsername(String username) {
        this.username = username;
    }
    public String getPassword() {
        return password;
    }
    public void setPassword(String password) {
        this.password = password;
    }
    public String getAddress() {
        return address;
    }
    public void setAddress(String address) {
        this.address = address;
    }
    public String getContact() {
        return contact;
    }
    public void setContact(String contact) {
        this.contact = contact;
    }
}

```

EmployeeServlet.java

```
package com.pu;
```

```
import java.sql.*;
```

```
import jakarta.servlet.*;
```

```
import java.io.*;
```

```
public class EmployeeServlet extends HttpServlet {
```

```

    protected void doPost(HttpServletRequest request, HttpServletResponse response) throws
    ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out=response.getWriter();
        int id = Integer.valueOf(request.getParameter("id"));
        String fn=request.getParameter("firstName");
    }
}

```

```

String ln=request.getParameter("lastName");
String un=request.getParameter("username");
String pwd=request.getParameter("password");
String addr=request.getParameter("address");
String cont=request.getParameter("contact");
Employee emp = new Employee();
emp.setId(id);
emp.setFirstName(fn);
emp.setLastName(ln);
emp.setUsername(un);
emp.setPassword(pwd);
emp.setAddress(addr);
emp.setContact(cont);
request.setAttribute("emp",emp);
int result = 0;
try{
    Class.forName("com.mysql.cj.jdbc.Driver");

    Connection connection =
DriverManager.getConnection("jdbc:mysql://localhost:3306/god","root","admin123");
    String query = "INSERT INTO employee1 (id, first_name, last_name, username, password,
address, contact) VALUES (?, ?, ?, ?, ?,?,?);";
    PreparedStatement preparedStatement = connection.prepareStatement(query);
preparedStatement.setInt(1, emp.getId());
preparedStatement.setString(2, emp.getFirstName());
preparedStatement.setString(3, emp.getLastName());
preparedStatement.setString(4, emp.getUsername());
preparedStatement.setString(5, emp.getPassword());
preparedStatement.setString(6, emp.getAddress());
preparedStatement.setString(7, emp.getContact());
    System.out.println(preparedStatement);
result = preparedStatement.executeUpdate();
    RequestDispatcher rd=request.getRequestDispatcher("employeedetail.jsp");
rd.forward(request,response);

    } catch (Exception e) {
        System.err.println(e);
    }
}
}

```

employeeregister.jsp

```

<%@page language="java" contentType="text/html; charset=ISO-8859-1"
pageEncoding="ISO-8859-1"%>
<!DOCTYPEhtml>
<html>
<head>
<metacharset="ISO-8859-1">

```

```

<title>Insert title here</title>
</head>
<body>
<h1>Employee Register Form</h1>
<formaction="EmployeeServlet"method="post">
<tablestyle="width: 50%">
<tr>
<td>Registration Id</td>
<td><inputtype="text"name="id"/></td>
</tr>
<tr>
<td>First Name</td>
<td><inputtype="text"name="firstName"/></td>
</tr>
<tr>
<td>Last Name</td>
<td><inputtype="text"name="lastName"/></td>
</tr>
<tr>
<td>UserName</td>
<td><inputtype="text"name="username"/></td>
</tr>
<tr>
<td>Password</td>
<td><inputtype="password"name="password"/></td>
</tr>
<tr>
<td>Address</td>
<td><inputtype="text"name="address"/></td>
</tr>
<tr>
<td>Contact No</td>
<td><inputtype="text"name="contact"/></td>
</tr>
</table>
<inputtype="submit"value="Submit"/></form>
</body>
</html>

```

employeedetail.jsp

```

<%@pagelanguage="java"contentType="text/html; charset=ISO-8859-1"
pageEncoding="ISO-8859-1"%>
<%@pageimport="com.pu.*"%>
<!DOCTYPEhtml>
<html>
<head>
<metacharset="ISO-8859-1">
<title>Insert title here</title>
</head>
<body>

```

```
<%  
    out.print("You are successfully registered");  
%>  
</body>  
</html>
```

RESULT:

Thus, to implement a simple MVC web application using JSP and Servlet.